Karatsuba → work: speed increase of multiplication

$$\begin{array}{r} 123 \\ \times\ 45 \\ \hline 615 \\ 4920 \\ \hline \end{array}$$

$O(n^2)$ — slow

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for( ){
    for( ){
    }
}
```

$$\begin{array}{r} 1234 \\ \times\ 5678 \\ \hline \end{array} \longrightarrow \begin{array}{r} a\ b \\ \times\ c\ d \\ \hline \end{array}$$

$a = 1200 = 12 \times 10^{n/2}$

$b = 34$

$c = 5600 = 56 \times 10^{n/2}$

$d = 78$

$$\begin{array}{r} a_{10^{n/2}}\ b_{10^0} \\ \times\ c_{10^{n/2}}\ d_{10^0} \\ \hline \end{array}$$

$a \times 10^{n/2} \cdot c \times 10^{n/2} + b \times c \times 10^{n/2} + a \times 10^{n/2} \cdot d + bd$

$= ac \times 10^n + bc \times 10^{n/2} + ad \times 10^{n/2} + bd$

$= ac \times 10^n + (bc + ad) \times 10^{n/2} + bd$

$= ac \times 10^n + (bc + ad + ac - ac + bd - bc) \times 10^{n/2} + bd$

$= ac \times 10^n + (a(c+d) + b(c+d) - ac - bd) \times 10^{n/2} + bd$

$= ac \times 10^n + ((a+b)(c+d) - ac - bd) \times 10^{n/2} + bd$

- $T(n) = 3T\left(\frac{n}{2}\right) + O(n)$ → Recursive relation [Multiply Three $n/2$ part divide]

  Recursive Relation

  $O(n^{1.58})$

  Time complexity

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simulate Karatsuba [Decimal number system]

$x \to a\ b$

$y \to c\ d$

step 1: $x = a \times 10^{n/2} + b$

$y = c \times 10^{n/2} + d$

step 2: Multiply,

$ac \times 10^n + ad \times 10^{n/2} + bc \times 10^{n/2} + bd$

$= ac \times 10^n + (ad + bc) \times 10^{n/2} + bd$

$= ac \times 10^n + (ad + bc + ac - ac + bd - bd) \times 10^{n/2} + bd$

$= ac \times 10^n + (a(d+c) + b(d+c) - ac - bd) \times 10^{n/2} + bd$

$= ac \times 10^n + ((a+b)(d+c) - ac - bd) \times 10^{n/2} + bd$

* Karatsuba can be done for hexadecimal, octal binary number systems too.

simulate Karatsuba [Hexadecimal number system]

$x \to a\ b$
$y \to c\ d$

step 1 : $x = a \times 16^{n/2} + b$
$y = c \times 16^{n/2} + d$

step 2 : Multiply,

$$ac \times 16^n + ad \times 16^{n/2} + bc \times 16^{n/2} + bd$$

$$= ac \times 16^n + (ad+bc) \times 16^{n/2} + bd$$

$$= ac \times 16^n + (ad+bc+ac-ac+bd-bd) \times 16^{n/2} + bd$$

$$= ac \times 16^n + (a(d+c) + b(d+c) - ac - bd) \times 16^{n/2} + bd$$

$$= ac \times 16^n + ((a+b)(d+c) - ac - bd) \times 16^{n/2} + bd$$

simulate Karatsuba [Binary number system]

$x \to a\ b$
$y \to c\ d$

step 1 : $x = a \times 2^{n/2} + b$
$y = c \times 2^{n/2} + d$

step 2 : Multiply,

$$ac \times 2^n + ad \times 2^{n/2} + bc \times 2^{n/2} + bd$$

$$= ac \times 2^n + (ad+bc) \times 2^{n/2} + bd$$

$$= ac \times 2^n + (ad+bc+ac-ac+bd-bd) \times 2^{n/2} + bd$$

$$= ac \times 2^n + (a(d+c) + b(d+c) - ac - bd) \times 2^{n/2} + bd$$

$$= ac \times 2^n + ((a+b)(d+c) - ac - bd) \times 2^{n/2} + bd$$